INTERNSHIP REPORT

ON

DATA SCIENCE MASTER VIRTUAL INTERNSHIP

PROJECT: HOUSE PRICE PREDICTION USING MACHINE LEARNING

Submitted To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

Bachelor of Technology in Computer Science and Engineering

Submitted By: PRAMOD KUMAR

Roll Number: 2410030367

College Email ID: pramod.kumar.cs28@iilm.edu

Batch: 2026-27

Internship: Data Science Master Virtual Internship

Duration: 8 Weeks | April - June 2026

Internship certificate issued under the AICTE- EduSkills Virtual Internship programme.

CANDIDATE’S DECLARATION

I, PRAMOD KUMAR, hereby declare that the internship report titled “Data Science Master Virtual Internship – House Price Prediction Using Machine Learning” has been prepared by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering.

I confirm that the work presented in this report is based on the project work completed during the internship period and that the sources used in preparing this report have been acknowledged appropriately. I further declare that this report has not been submitted to any other institute or university for the award of any degree or diploma and that I shall be responsible for any copyright or academic-integrity violation arising from the submission.

Student Name: PRAMOD KUMAR

Roll No.: _2410030367

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to everyone who supported me during the completion of the Data Science Master Virtual Internship and the preparation of this report. The internship provided an opportunity to apply data science and machine learning concepts in a practical project environment.

I am thankful to the School of Computer Science and Engineering, IILM University, Greater Noida, for providing the academic environment and support required for completing this work. I also acknowledge the internship programme ecosystem associated with AICTE, EduSkills and the National Internship Portal. The internship certificate identifies Siemens as the supporting organization.

I am grateful to my faculty members, mentors, parents and friends for their guidance, encouragement and continuous support throughout the internship and project work.

Student Name: PRAMOD KUMAR

Roll No.: _2410030367

INTERNSHIP COMPLETION CERTIFICATE

The following certificate was issued for successful completion of the 8-week Data Science Master Virtual Internship during April- June 2026. The certificate identifies the student as PRAMOD KUMAR, IILM University, Greater Noida, and shows AICTE, EduSkills and the Ministry of Education/National Internship Portal branding, with Siemens listed as the supporting organization.

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Certificate of Virtual Internship

This is to certify that

Pramod kumar

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Science Master Virtual Internship

During April - June 2026

May this Internship learning propel you toward a bright and successful career.

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Certificate ID : 4cf28d21fde5fe6d9d8d

Student ID: STU699acedf75b401771753183

INTERNSHIP GRADE O

GRADE- O (Outstanding):90-100 | E (Excellent):80-89 | A (Very Good):70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

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4. PROJECT DESCRIPTION

4.1 Introduction

The Data Science Master Virtual Internship was completed as an 8-week virtual internship during April- June 2026. The project presented in this report focuses on applying data science and machine learning techniques to the problem of house price prediction.

House price prediction is a regression problem in which historical property information is used to estimate the price of a property. The project uses an India housing-price dataset and develops a complete machine learning workflow covering data inspection, preprocessing, outlier treatment, categorical encoding, feature scaling, model training, evaluation and comparison.

The final workflow evaluates six regression algorithms: Linear Regression, Ridge Regression, Lasso Regression, Decision Tree, Random Forest and Gradient Boosting. On the recorded test split, Random Forest produced the strongest performance, with an R^2 Score of 0.9957, MAE of 7.26 and RMSE of 9.20.

4.2 Organization / Internship Programme Profile

The supplied internship certificate identifies the programme as the "Data Science Master Virtual Internship". It states that PRAMOD KUMAR of IILM University, Greater Noida successfully completed an 8-week internship during April- June 2026. The certificate carries Ministry of Education, AICTE National Internship Portal and EduSkills branding and lists Siemens as the supporting organization.

Programme Detail	Certificate-Supported Information
Internship Programme	Data Science Master Virtual Internship

Programme Detail	Certificate-Supported Information
Duration	8 Weeks
Period	April – June 2026
Student	PRAMOD KUMAR | Roll No. 2410030367 | pramod.kumar.cs28@iilm.edu
Institution	IILM University, Greater Noida
Programme Ecosystem	AICTE – EduSkills Virtual Internship
Supporting Organization	Siemens
Certificate	Certificate of Virtual Internship

Note: The certificate does not provide a detailed organizational profile, mentor name, project-task list or office/department details. Therefore, these details have not been invented in this report.

4.3 Problem Statement

Property prices are affected by several numerical and categorical characteristics such as location, property size, number of bedrooms, property type, building information and accessibility-related factors. Manual estimation of prices can be difficult when many variables must be considered simultaneously.

The problem addressed by this project is to build a machine learning regression system that learns patterns from historical Indian housing data and predicts the Price_in_Lakhs value of a property. A second objective is to compare different regression algorithms using standard evaluation metrics and identify the best-performing model.

4.4 Project Objectives

- To understand and prepare a large housing-price dataset for machine learning.
- To inspect the dataset structure, data types and missing values.
- To handle missing values through median and mode-based preprocessing.
- To identify and remove numerical outliers using the Interquartile Range (IQR) method.
- To convert categorical information into numerical features.
- To standardize the prepared feature set using StandardScaler.
- To train and compare six regression models.
- To evaluate models using R^2 Score, Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE).
- To identify the best-performing regression model for the supplied dataset.
- To visualize model performance, prediction quality, residuals, feature importance, correlations and price distribution.

4.5 Scope of the Project

The scope of this project is the development and evaluation of a machine learning pipeline for house-price prediction using the supplied Indian housing dataset.

- Data inspection and preprocessing.
- Handling missing values and numerical outliers.
- Encoding categorical variables.
- Feature scaling and train-test splitting.
- Regression model training and comparison.
- Evaluation using R^2, MAE and RMSE.
- Visualization of model performance and data relationships.
- Identification of a suitable model for future deployment.

The supplied project material does not demonstrate production deployment, a live data feed or a user-facing application. Such features can be treated as future extensions rather than completed internship deliverables.

4.6 Technologies and Tools Used

Technology / Tool	Purpose
Python	Programming and implementation of the data science workflow.
NumPy	Numerical operations and array-based computation.
Pandas	Data loading, inspection, cleaning and tabular manipulation.
Matplotlib	Data and model-performance visualization.
Seaborn	Statistical visualization and correlation analysis.
Scikit-learn	Preprocessing, splitting, regression models and evaluation metrics.
Jupyter / Notebook Environment	Development and execution of the machine learning workflow.

Table 4.1 — Technologies and Tools Used

4.7 System Architecture

The project follows a sequential machine learning pipeline. The raw housing dataset is loaded into a Pandas DataFrame, inspected and cleaned. Numerical outliers are treated using the IQR method. Categorical attributes are encoded, the target variable is separated, features are standardized and the data is split into training and testing sets. Six regression models are then trained and evaluated, after which the best-performing model is selected.

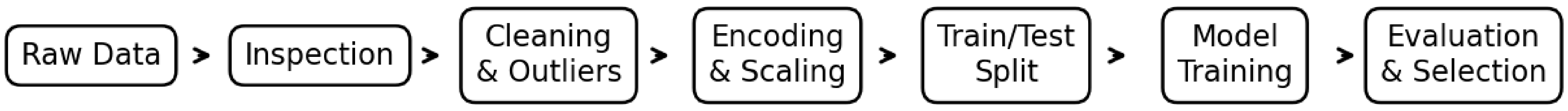

Figure 4.1 — End-to-End Machine Learning Architecture

Stage	Input	Processing	Output
1. Data Loading	CSV housing dataset	Read using Pandas	DataFrame
2. Data Inspection	DataFrame	Shape, info, statistics and missing-value checks	Data-quality information
3. Data Cleaning	Raw records	Median/mode preprocessing	Clean dataset
4. Outlier Removal	Numerical features	IQR-based filtering	Reduced-noise dataset
5. Encoding	Categorical features	Label encoding / dummy variables	Model-ready features
6. Scaling&Split	Features + target	StandardScaler + 80/20 split	Training/testing sets
7. Model Training	Training set	Six regression algorithms	Trained models
8. Evaluation	Testing set	R^2, MAE and RMSE	Performance comparison
9. Selection	Model scores	Performance-based selection	Random Forest selected

Table 4.2 — System Architecture Stages

4.8 Methodology

4.8.1 Dataset Description

The project uses the dataset file "india_housing_prices.csv" . The dataset contains 250,000 property records and 23 columns. The prediction target is Price_in_Lakhs. The available attributes cover identification, location, property characteristics, building information, nearby facilities, accessibility, parking, security, amenities, facing, ownership and availability-related information.

Item	Value
Original records	250,000
Original columns	23
Target variable	Price_in_Lakhs
Records after IQR outlier removal	229,980
Training records after preprocessing	183,984
Testing records after preprocessing	45,996
Input features used for prediction	22

Table 4.3 — Dataset Summary

4.8.2 Data Inspection

The workflow begins by importing NumPy, Pandas, Matplotlib and Seaborn and loading the housing dataset. Dataset shape, initial records, data types and summary information are inspected. A missing-value analysis is performed before applying the preprocessing steps.

4.8.3 Handling Missing Values

The preprocessing workflow checks missing values for each column. The recorded dataset initially contains no missing values. The notebook nevertheless includes a general cleaning approach in which numerical columns can be filled using their median and categorical columns using their mode. After preprocessing, the workflow reports zero missing values.

4.8.4 Outlier Removal

Numerical outliers are treated using the Interquartile Range method. Q1 represents the 25th percentile and Q3 represents the 75th percentile, while IQR is calculated as Q3 − Q1. Values below Q1 − 1.5× IQR or above Q3 + 1.5 × IQR are treated as outliers. The dataset is reduced from 250,000 to 229,980 records after this filtering.

4.8.5 Categorical Encoding and Feature Preparation

Categorical attributes are converted into numerical representations using encoding techniques. The workflow uses LabelEncoder and also uses get_dummies() for the final model-ready feature matrix. Price_in_Lakhs is separated as the target variable, while the remaining prepared attributes form the input feature matrix.

4.8.6 Feature Scaling

StandardScaler is used to standardize the feature matrix. Scaling places numerical features on a comparable scale and creates a consistent feature representation for the machine learning models.

4.8.7 Train-Test Split

The processed dataset is divided into training and testing sets using an 80:20 split with random_state=42. The recorded resulting training set contains 183,984 samples and the testing set contains 45,996 samples, with 22 input features.

4.8.8 Machine Learning Models

Model	Description

Model	Description
Linear Regression	Baseline linear regression model.
Ridge Regression	Linear regression with L2 regularization.
Lasso Regression	Linear regression with L1 regularization.
Decision Tree	Tree-based model that learns decision rules.
Random Forest	Ensemble model built from multiple decision trees.
Gradient Boosting	Sequential boosting-based regression model.

Table 4.4 — Regression Models Used

4.8.9 Evaluation Metrics

Three evaluation metrics are used. R^2 Score indicates how much of the variation in the target is explained by the model; a higher value indicates better performance. Mean Absolute Error (MAE) measures the average absolute prediction error. Root Mean Squared Error (RMSE) gives greater weight to larger errors. Lower MAE and RMSE indicate lower prediction error.

4.8.10 Model Evaluation Results

Model	R^2 Score	MAE	RMSE
Random Forest	0.9957	7.26	9.20
Decision Tree	0.9916	9.59	12.78
Gradient Boosting	0.9901	11.32	13.91
Linear Regression	0.7379	56.18	71.53
Ridge Regression	0.7379	56.18	71.53
Lasso Regression	0.7379	56.17	71.53

Table 4.5 — Model Evaluation Results

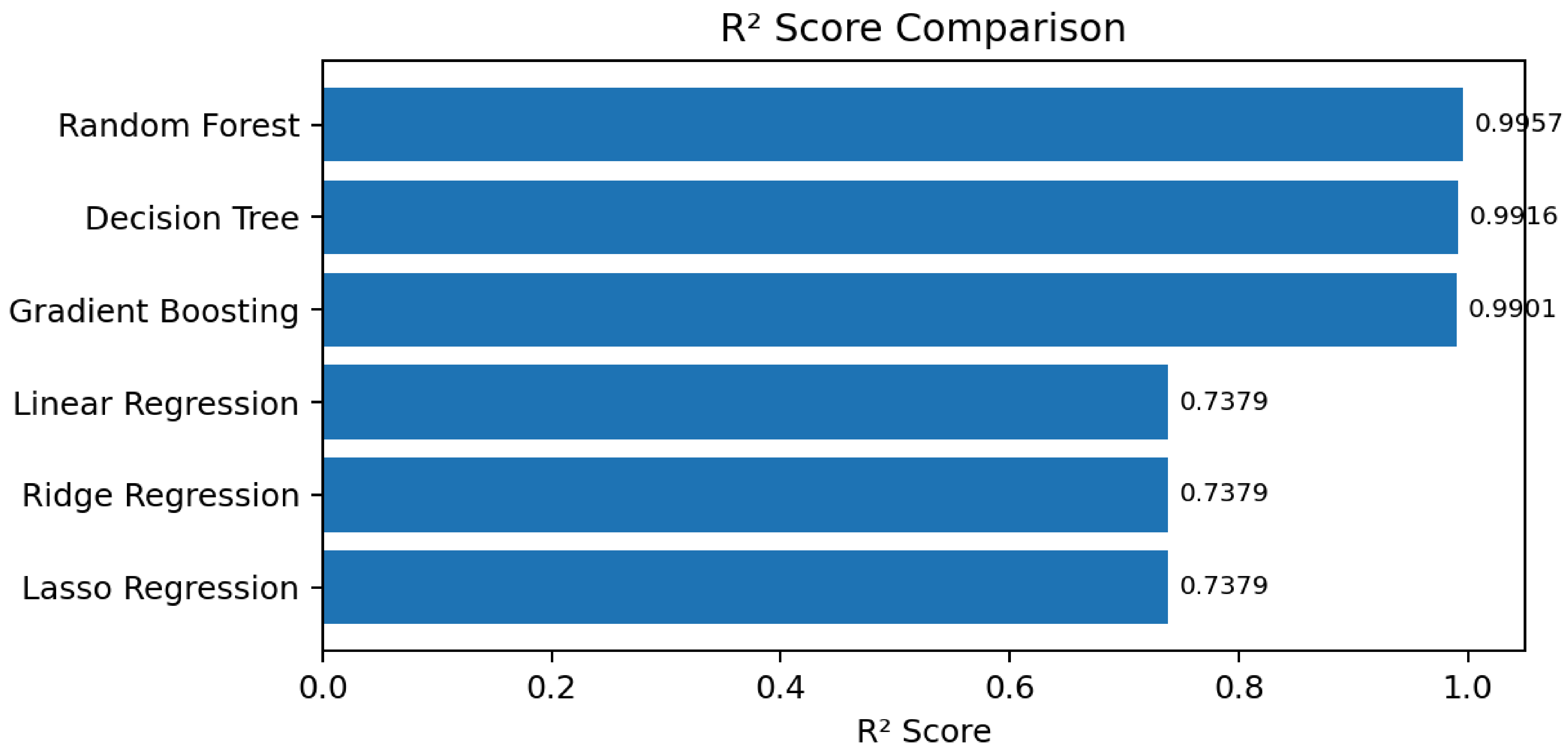

Figure 4.2 — R^2 Score Comparison of Regression Models

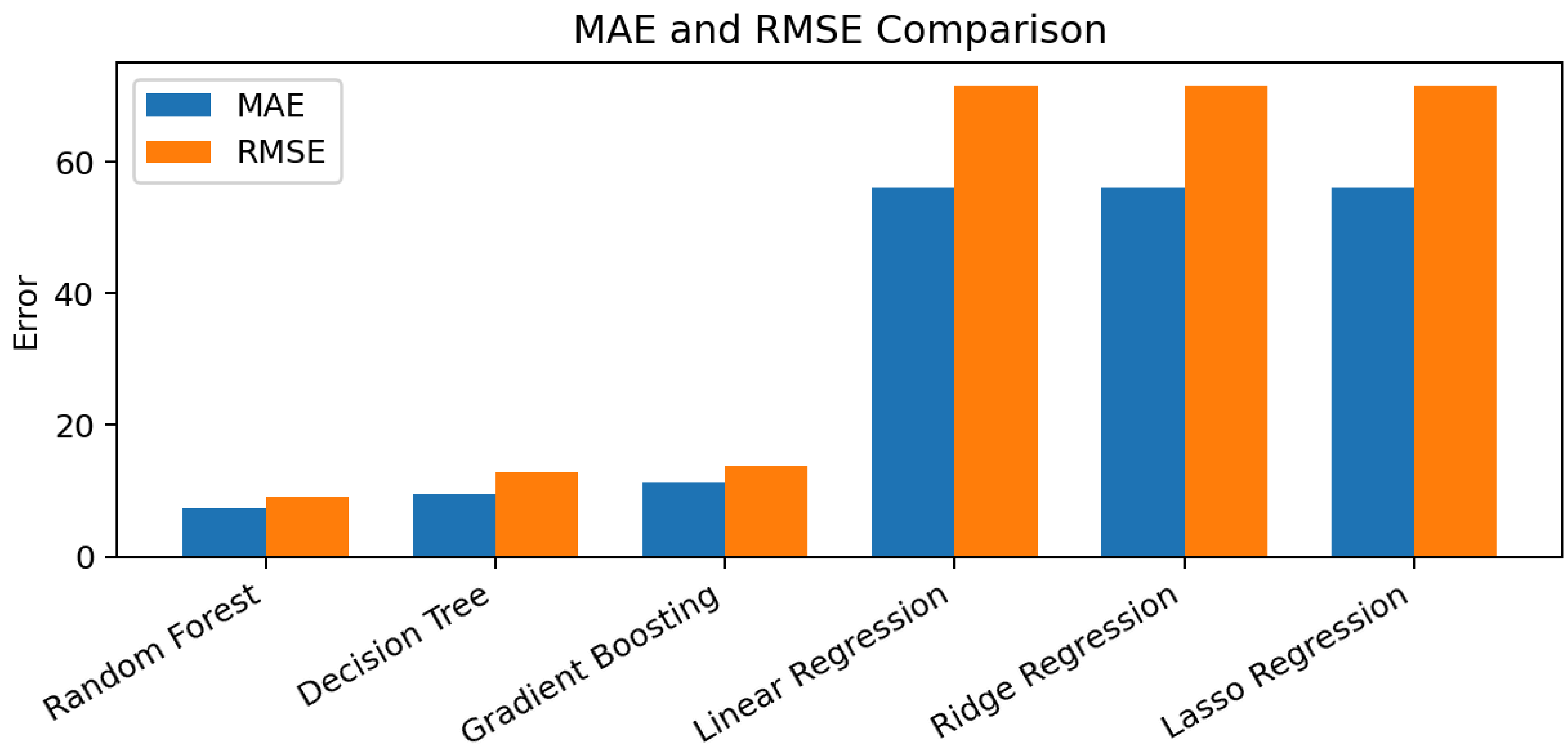

Figure 4.3 — Error Comparison of Regression Models

4.8.11 Interpretation of Results

The recorded results show that Random Forest is the strongest model among the six tested algorithms. It provides the highest R^2 Score and the lowest MAE and RMSE in the reported comparison. Decision Tree and Gradient Boosting also perform strongly, while Linear Regression, Ridge Regression and Lasso

Regression have substantially lower R^2 values and larger errors on the recorded test set.

The result demonstrates the usefulness of non-linear tree-based ensemble methods for this dataset. The Random Forest model can capture complex relationships between property characteristics and price that are not represented as effectively by the tested linear baselines.

4.8.12 Visualization and Analysis

- R^2 Score Comparison — compares predictive performance across all tested models.
- Actual vs Predicted Plot — compares actual prices with predictions from the best model.
- Residuals Plot — examines prediction errors against predicted values.
- Feature Importance — identifies influential features using the Random Forest model.
- Correlation Heatmap — visualizes relationships among numerical variables.
- Price Distribution — examines the distribution of Price_in_Lakhs and its log-transformed distribution.

4.8.13 Best Model

The Random Forest Regressor is identified as the best-performing model in the project. Its recorded test performance is R^2 = 0.9957, MAE = 7.26 and RMSE = 9.20. The model is an ensemble of decision trees and provides strong performance for the supplied housing-price dataset.

4.9 Expected Outcomes

The project provides an end-to-end machine learning workflow for estimating house prices from property attributes and demonstrates the effect of data preprocessing, model selection and evaluation on prediction quality.

- A cleaned and model-ready housing dataset.
- A reproducible preprocessing and model-training workflow.
- Performance comparison of six regression algorithms.
- Identification of Random Forest as the best-performing model on the supplied test split.
- Quantitative evaluation using R^2, MAE and RMSE.
- Visual analysis of model performance and data relationships.
- A foundation for future deployment as a user-facing house-price prediction application.

4.10 Certificates of Completion and Communication Mail Proof

The internship report format requires the internship completion certificate and relevant communication/e-mail proof to be attached. The supplied certificate has been included in Chapter 3 of this report.

Additional official communication proof, if required by the university, should be inserted below or added as an appendix without modifying the original document.

Communication / E-mail Proof: [ATTACH OFFICIAL COMMUNICATION HERE]

PROJECT CONCLUSION

The Data Science Master Virtual Internship project provided an opportunity to apply a complete machine learning workflow to house-price prediction. The project covered data inspection, preprocessing, outlier treatment, categorical encoding, feature scaling, train-test splitting, model training and evaluation.

Six regression algorithms were compared using R^2 Score, MAE and RMSE. The recorded results identify Random Forest as the best-performing model, achieving an R^2 Score of 0.9957, MAE of 7.26 and RMSE of 9.20. The project demonstrates how data science and machine learning can be applied to structured real-estate data and provides a foundation for future work such as deployment, hyperparameter tuning, explainable AI and a user-friendly prediction interface.

5. BIBLIOGRAPHY / REFERENCES

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Matplotlib Development Team. Matplotlib Documentation.

Waskom, M. Seaborn: Statistical Data Visualization — Documentation.

Project notebook: “house-price-prediction-using-machinew-learn-977934.ipynb”.

Internship report format: “Internship Report Format updated 26-27.docx”.

Certificate of Virtual Internship: “Data Science Master Virtual Internship”, April- June 2026.

APPENDIX A — INTERNSHIP OFFER LETTER

The following pages reproduce the internship offer letter supplied with this report. The letter confirms the student details, internship domain, duration and the 8-week AICTE- EduSkills programme.

Student: PRAMOD KUMAR | Roll No.: 2410030367 | College Email: pramod.kumar.cs28@iilm.edu

Ref No: C16Y2026M051398417

Internship Offer Letter

Student Name	: Pramod kumar
Stream	: B.Tech
Branch	: B.Tech(Computer Science and Engineering (Cyber Security))
Institute Name	: IILM University, Greater Noida
Internship Domain	: Siemens Data Science Master
Internship Duration	: April 2026 - June 2026
AICTE Student ID	: STU699acedf75b401771753183

We are pleased to inform you that you have been selected for the 8-Week **AICTE – EduSkills Virtual Internship Program**.

During this internship, you will learn and demonstrate industry-relevant skills that will enhance your employability and strengthen your confidence in the subject area.

The program is designed to provide structured learning combined with practical exposure.

Internship Structure & Evaluation:

- The internship will follow a **week-wise structured learning plan**.
- You will be required to complete **weekly assessments** to track your progress and understanding.
- Certain modules will require submission of **project documentation and assignments**.
- In the final week, you must appear for a **Final Assessment Test**.
- The final grade will be based on your overall performance, including weekly assessments, project submissions, and the final test.

Upon successful completion of all required steps and assessments, you will be awarded a **Virtual Internship Certificate**.

Please note that this offer letter will be considered valid only after the successful completion of the internship program and fulfillment of all assessment requirements, including obtaining the final completion certificate. Without the final completion certificate, this offer letter shall not be treated as a valid document for any official purpose.

The week wise Structure of the Internship is as Follows:

Week	Module	Description
Week 1	Applications & Use Cases Professional	Learn to identify business problems suitable for AI and machine learning.
Week 2	Data Engineering Professional	Learn to access, load, transform, and calculate data.
Week 3	Machine Learning Professional	Build foundational machine learning skills.
Week 4	Data Preparation Master	Master advanced data handling and automation.
Week 5	Machine Learning Master	Learn advanced machine learning techniques.
Week 6	Applications & Use Cases Master	Deploy and monitor ML models.
Week 7	Platform Administration Master	Configure and manage RapidMiner AI Hub.
Week 8	Platform Administration Master	Enterprise-level platform operations.
Final Credential Validation / Internship Grade Point Assessment		

Note: There is no stipend associated with this internship.

We congratulate you on your selection and wish you a successful learning journey.

Bibek Ranjan
Director, EduSkills

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APPENDIX B — VERIFIED INTERNSHIP DETAILS

Detail	**Information**
Reference Number	C16Y2026M051398417
Student Name	Pramod kumar
Stream	B.Tech
Branch	B.Tech (Computer Science and Engineering (Cyber Security))
Institute	IILM University, Greater Noida
Internship Domain	Siemens Data Science Master
Internship Duration	April 2026 – June 2026
AICTE Student ID	STU699acedf75b401771753183
College Email ID	pramod.kumar.cs28@iilm.edu
Roll Number	2410030367

The offer letter states that the internship is an 8-week AICTE- EduSkills Virtual Internship and includes weekly learning, assessments, project documentation/assignments and a final assessment. It also states that there is no stipend associated with the internship.